

4 (a) (i)  $T = 0.60 \text{ s}$  and  $\omega = 2\pi/T$  C1

$$\omega = 10(10.47) \text{ rad s}^{-1}$$

A1 [2]

(ii) energy =  $\frac{1}{2}m\omega^2x_0^2$  or  $\frac{1}{2}mv^2$  and  $v = \omega x_0$  C1

$$= \frac{1}{2} \times 120 \times 10^{-3} \times (10.5)^2 \times (2.0 \times 10^{-2})^2$$

$$= 2.6 \times 10^{-3} \text{ J}$$

A1 [2]

(b) sketch: smooth curve in correct directions B1

peak at  $f$  M1

amplitude never zero and line extends from  $0.7f$  to  $1.3f$  A1 [3]

(c) sketch: peaked line always below a peaked line A M1

peak not as sharp and at (or slightly less than) frequency of peak in line A A1 [2]

|               |                                                           |                 |              |
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